PROBLEM STATEMENT

Commercial Business Intelligence (BI) platforms such as Microsoft Power BI, Tableau, and Zoho Analytics impose steep recurring subscription fees, strict user-licensing restrictions, and mandatory cloud data custody.

Organizations, academic institutions, and privacy-conscious enterprises often handle sensitive proprietary data that cannot be entrusted to third-party clouds. Existing open-source alternatives are often fragmented, difficult to deploy, and lack intuitive drag-and-drop dashboard design, self-contained ETL pipelines, or lightweight zero-cost infrastructure.

OUR SOLUTION

GYANS Analytics is an open-source, self-hosted BI platform that enables users to connect data sources, clean and transform datasets, and construct interactive dashboards via an intuitive drag-and-drop canvas.

The system eliminates SaaS subscription overheads, maintains absolute on-premise data sovereignty, and delivers fast, real-time visual analytics for modern teams.

• Basic Workflow

USER / ANALYST
↓
Upload Dataset (CSV / Excel / SQL DB)
↓
Automated Schema Detection & Preview
↓
Visual Query Builder & Metric Aggregation
↓
Drag-and-Drop Widget Configuration
↓
Interactive Dashboard Layout Creation
↓
Dynamic Multi-chart Filtering & Drill-down
↓
Export Reports (PDF/PNG) & Embed Securely

KEY FEATURES

Data Analyst

- Multi-format ingestion (CSV, Excel, JSON)
- Direct SQL database connectivity
- Drag-and-drop dashboard composer
- 15+ rich chart widgets (ECharts)
- Custom SQL query editor with live preview
- Data transformation & calculated fields

Team Collaborator

- Interactive shared dashboard viewing
- Dynamic cross-chart filtering & date pickers
- Export dashboards to PDF & PNG formats
- Embeddable widgets via iframe & tokens
- Dark & Light mode visual theme customization
- Scheduled automated report distributions

Administrator

- Role-Based Access Control (Admin / Editor / Viewer)
- Multi-workspace isolation & member governance
- Encrypted connection credentials storage
- Query caching & performance optimization
- Audit logging & user activity monitoring
- One-command Dockerized deployment

TECHNOLOGY STACK

Layer	Technology
Frontend Framework	React.js 18 + TypeScript + Vite
UI & Design System	Ant Design 5 (Enterprise Components)
Visualization Engine	Apache ECharts + Recharts (15+ Chart Widgets)
Backend Framework	Python 3.12 + FastAPI (Asynchronous REST API)
Data Processing & ORM	Pandas + SQLAlchemy 2.0 (Async) + Alembic
Database Engine	SQLite (Development) / PostgreSQL 16 (Production)
Authentication & Security	OAuth2 + JWT (JSON Web Tokens) + Bcrypt Hashing
DevOps & Containerization	Docker + Docker Compose + Nginx Reverse Proxy

FUTURE ENHANCEMENTS

- Natural Language Querying (NLQ) powered by local LLMs (Ollama / Mistral) for conversational analytics
- Automated AI Insights, time-series trend forecasting, and anomaly detection (Prophet / Scikit-learn)
- Cloud Lakehouse & Warehouse Connectors (Google BigQuery, Snowflake, and AWS Redshift)
- Visual node-based ETL pipeline canvas for drag-and-drop data blending and automated joins
- Mobile-first Progressive Web App (PWA) view with touch-optimized KPI cards